

Assignment No	: 1
Roll No	: 381
Batch	: E4
PRN No	: 202502040032
Title of the Assignment	: Core Java Fundamentals

a) Write a program using Exception Handling, Abstraction, and Package

Solution : Smart vehicle rental system .

Directory Structure :
<pre>src ├── abstractdemo │ ├── Bike.java │ └── Vehicle.java ├── exceptiondemo │ ├── NoFuelExc.java │ └── Vehicle.java ├── interfacedemo │ ├── Abc.java │ └── Xyz.java └── mainapp └── MainApp.java</pre>
abstractdemo/Bike.java
<pre>package abstractdemo; public class Bike extends Vehicle { public void startRide() { System.out.println("Bike ride started for customer"); } }</pre>
abstractdemo/Vehicle.java
<pre>package abstractdemo; public abstract class Vehicle { public abstract void startRide();</pre>

```
public void stopRide() {  
    System.out.println("Ride completed. Vehicle parked safely");  
}  
}
```

exceptiondemo/NoFuelExc.java

```
package exceptiondemo;  
  
public class NoFuelExc extends Exception {  
    public NoFuelExc(String msg) {  
        super(msg);  
    }  
}
```

exceptiondemo/Vehicle.java

```
package exceptiondemo;  
  
public class Vehicle {  
  
    public void checkFuelAndStart(int fuel) throws NoFuelExc {  
        if (fuel <= 0) {  
            throw new NoFuelExc("Booking Failed: Vehicle has no fuel");  
        } else {  
            System.out.println("Vehicle ready for service. Fuel = " + fuel + " litres");  
        }  
    }  
}
```

interfacedemo/Abc.java

```
package interfacedemo;  
  
public interface Abc {  
    void pay(double amount);  
    void generateReceipt();  
}
```

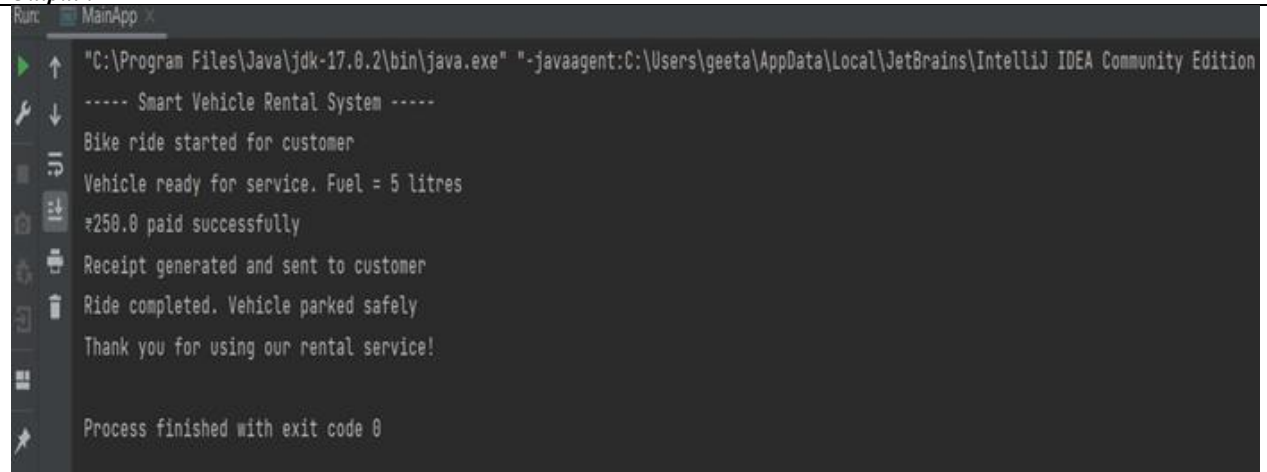
interfacedemo/Xyz.java

```
package interfacedemo;  
  
public class Xyz implements Abc {  
    public void pay(double amount) {  
        System.out.println("₹" + amount + " paid successfully");  
    }  
  
    public void generateReceipt() {
```

```
        System.out.println("Receipt generated and sent to customer");  
    }  
}
```

mainapp/MainApp.java

```
package mainapp;  
  
import exceptiondemo.Vehicle;  
import exceptiondemo.NoFuelExc;  
import interfacedemo.Xyz;  
import abstractdemo.Bike;  
  
public class MainApp {  
  
    public static void main(String[] args) {  
  
        System.out.println("----- Smart Vehicle Rental System -----");  
  
        Bike bike = new Bike();  
        bike.startRide();  
  
        Vehicle service = new Vehicle();  
  
        try {  
            service.checkFuelAndStart(5);  
        } catch (NoFuelExc e) {  
            System.out.println(e.getMessage());  
            return;  
        }  
  
        Xyz payment = new Xyz();  
        payment.pay(250);  
        payment.generateReceipt();  
  
        bike.stopRide();  
  
        System.out.println("Thank you for using our rental service!");  
    }  
}
```

Output :

```
Run: MainApp x  
"C:\Program Files\Java\jdk-17.0.2\bin\java.exe" "-javaagent:C:\Users\geeta\AppData\Local\JetBrains\IntelliJ IDEA Community Edition  
----- Smart Vehicle Rental System -----  
Bike ride started for customer  
Vehicle ready for service. Fuel = 5 litres  
₹250.0 paid successfully  
Receipt generated and sent to customer  
Ride completed. Vehicle parked safely  
Thank you for using our rental service!  
  
Process finished with exit code 0
```

Submitted by,
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